HDOC Day-46 - C Programming

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Problem Statement

Write a C program that reads the size and elements of a variable-length integer array **nums[]**. Pass the array using pointers to **Bubble_sort()** and sort it in ascending order using Bubble Sort. Then pass the sorted array to **occurrence()**, which reads a target integer, finds its first and last occurrences, displays their values and indices, and prints **-1, -1** when the target is absent.

C Program

```
#include <stdio.h>

void Bubble_sort(int *nums, int n)
{
    int i, j, temp;
    for (i = 0; i < n - 1; i++)
    {
        for (j = 0; j < n - i - 1; j++)
        {
            if (*(nums + j) > *(nums + j + 1))
            {
                temp = *(nums + j);
                *(nums + j) = *(nums + j + 1);
                *(nums + j + 1) = temp;
            }
        }
    }
}

void occurrence(int *nums, int n)
{
    int target, first = -1, last = -1, i;

    printf("Enter target: ");
    scanf("%d", &target);

    for (i = 0; i < n; i++)
    {
        if (*(nums + i) == target)
        {
            if (first == -1)
                first = i;
            last = i;
        }
    }

    if (first == -1)
        printf("-1, -1\n");
    else
    {
        printf("First occurrence: %d at index %d\n",
               *(nums + first), first);
        printf("Last occurrence: %d at index %d\n",
               *(nums + last), last);
    }
}

int main()
{
    int n, i;

    printf("Enter size of array: ");
    scanf("%d", &n);

    int nums[n];
```

```
    printf("Enter %d elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &nums[i]);

    Bubble_sort(nums, n);

    printf("Sorted array: ");
    for (i = 0; i < n; i++)
        printf("%d ", nums[i]);
    printf("\n");

    occurrence(nums, n);

    return 0;
}
```

Execution / Output

